

Lecture 4: JavaScript Basics – The Logic of the Web

Goal: Master the fundamental "grammar" of JavaScript—the language that gives your website its brain.

- **1. Introduction: What is JavaScript?**
- If HTML is the **chassis** of a car and CSS is the **paint and leather seats**, then JavaScript is the **engine and electrical system**.
- Without JavaScript, a website is just a static page—like a printed newspaper. You can look at it, but you can't interact with it. JavaScript (JS) allows the page to "do" things: calculate math, respond to clicks, validate user input, and update content without reloading the page.

2. Variables: The "Storage Boxes"

In programming, we need to save information to use it later. We do this using **variables**. Think of a variable as a labeled storage box.

In modern JavaScript, we use two main keywords to create these boxes: `const` and `let`.

- **`const` (Constant):** Use this for data that **never changes**.
 - *Real-life example:* The date of your birth. Once set, it doesn't change.
 - `const birthYear = 2005;`
- **`let`:** Use this for data that **will change** later.
 - *Real-life example:* Your current bank balance. It goes up and down as you spend or earn.
 - `let bankBalance = 100;`

```
JavaScript    
  
// Example  
const pi = 3.14;           // We don't want this changing!  
let score = 0;           // The player's score will go up  
score = 10;             // We can change the value of 'let' later
```

3. Data Types: What's in the Box?

JavaScript needs to know what kind of information you are putting in the "box." Here are the three most important types for beginners:

Data Type	Description	Example
String	Text. Always wrapped in quotes.	"Hello World"
Number	Integers or decimals for math.	25 or 99.9
Boolean	A light switch: True or False.	true or false

Tip: If you put numbers inside quotes (like "25"), JavaScript treats it as **text** (String), not a number you can do math with!

4. Flow Control: Making Decisions

Websites need to be "smart." They need to make decisions based on user actions. This is called **Flow Control**.

A. Conditionals (**if** / **else**)

This is like a fork in the road. You tell the computer: *"If this condition is true, do this; otherwise, do that."*

```
JavaScript
let userAge = 18;

if (userAge >= 18) {
  console.log("Access Granted");
} else {
  console.log("Access Denied");
}
```

B. Loops (**for**)

What if you have to greet 100 people? You wouldn't write the code 100 times. You use a loop to tell the computer to repeat an action.

JavaScript

```
// This runs 5 times, printing the number each time
for (let i = 1; i <= 5; i++) {
  console.log("Hello, Guest #" + i);
}
```

5. Why does this matter for Frontend?

You might wonder, "Why do I need to learn logic if I just want to design websites?"

Here is why: **Everything you see on a modern website is logic.**

1. **Dynamic UI:** When you visit a profile page, the site uses a variable (the user's name) to display "Welcome, Ali" instead of just "Welcome."
2. **Form Validation:** When you sign up for an account, JavaScript uses an **if** statement to check: "Did the user leave the password box empty? If yes, show an error."
3. **Interactive Galleries:** When you click "Next" on a photo gallery, a loop or counter changes which image variable is displayed on the screen.

Key Takeaway: You aren't just memorizing commands. You are learning how to give instructions to a computer. Master these basics, and you will have the foundation to build any interactive interface you can imagine.